

Numerical Analysis PhD Exam, January 2017

You must show all your work as neatly and clearly as possible and indicate the final answer clearly. You may not use a calculator.

1. This problem has two parts.

- (a) Show that the equation $x = \frac{1}{2} \cos(x)$ has a solution α .
- (b) Find an interval $[a, b]$ containing α and such that for every $x_0 \in [a, b]$, the iterative sequence

$$x_{n+1} = \frac{1}{2} \cos x_n$$

will converge to α . Justify your answer.

2. For the basic Lagrange polynomials

$$L_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j} \quad \text{for } i = 0, \dots, n,$$

show that

$$\sum_i L_i(x) = 1 \quad \text{for all } x.$$

3. Consider a quadrature rule of the form for $0 < \alpha < 1$:

$$\int_0^1 x^\alpha f(x) dx \approx A \int_0^1 f(x) dx + B \int_0^1 x f(x) dx.$$

- (a) Determine the constants A, B so that the quadrature formula has maximum degree of exactness.
- (b) What is the degree of exactness of the formula in part (a)?

4. Let f be an arbitrary (continuous) function on $[0, 1]$ satisfying

$$f(x) + f(1 - x) \equiv 1 \quad \text{for } 0 \leq x \leq 1.$$

- (a) Show that $\int_0^1 f(x) dx = \frac{1}{2}$.
- (b) Show that the composite trapezoidal rule for computing $\int_0^1 f(x) dx$ is exact.

5. Assume that you are solving the initial value problem

$$\begin{aligned} y' &= f(t, y), \quad a \leq t \leq b, \\ y(a) &= \alpha. \end{aligned}$$

The formula for the global error of the numerical solutions for the ODE problem above obtained via Euler's method is ($M = \|Y''\|_\infty$):

$$|Y(t_i) - w_i| < \frac{hM}{2L} [e^{L(b-a)} - 1].$$

Compute the values of L and $M = \|Y''\|_\infty$ necessary to apply the global error formula above to the specific ODE problem

$$\begin{aligned} y' &= \sin(t + 2y) + e^t, \quad 0 \leq t \leq 1, \\ y(0) &= 0. \end{aligned}$$